Gossip Cell Devs

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Assignment 2

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Conceptual Model

The system modeled is the propagation of gossip within a population that is represented by a two-dimensional grid. Everyone occupies a cell in the grid and can either know or not know a piece of gossip.

At each simulation step, individuals interact with their neighbors and may acquire or lose gossip based on probabilities that are pre-defined. This system is representative of real-world information diffusion processes such as how rumors can spread in social networks or information in systems.

Model Overview

Each cells has a binary state of knows = 1 or knows = 0, where a 1 represents them knowing the gossip. There are two main types of neighborhoods that we are considered, Moore (8 surrounding cells) or Von Neumann (4 adjacent cells).

Formal Specifications

$\langle X,Y,S,N,d,\tau,\delta int,\delta ext,\lambda,D\rangle$

$S = \{knows\}$

Xlist = N/A (no external input)

Ylist = N/A (no external output)

$X = \{x \mid x \in s\}$

$Y = \{y \mid y \in s\}$

N(neighbors) = $\{(-1,0),(1,0),(0,-1),(0,1),(0,0)\}$ (Von Neumann),

$\{(i,j) | i,j \in \{-1,0,1\}\}$ (Moore)

d = 1 time unit

$T = N \rightarrow S$

Where n = number of neighbors that know the gossip

P = spread probability

K = forget probability.

The following rules exist.

If the cell does not know the gossip

$P(activation) = 1 – (1- P)^n$

λ, δext, δint All defined as per the specifications

Modifications From Old Model

There are many differences between the original model and the one developed in Cadmium. The core behavior and conceptual rules of gossip propagation remain the same, although I did make changes to implement it to the Cadmium framework.

Single Layer vs Two Layer Design

The most significant difference between the original model and the implemented version is the removal of the second layer. The original model uses a two-layer structure where the first layer represents the individuals and their gossip state, and the second layer represents the random values used to determine their behavior. In this implementation the model is refined to a single layer grid, where the randomness is generated directly within each cell using a pseudo random number generator. This modification reduces model complexity but preserves the logic of the random nature.

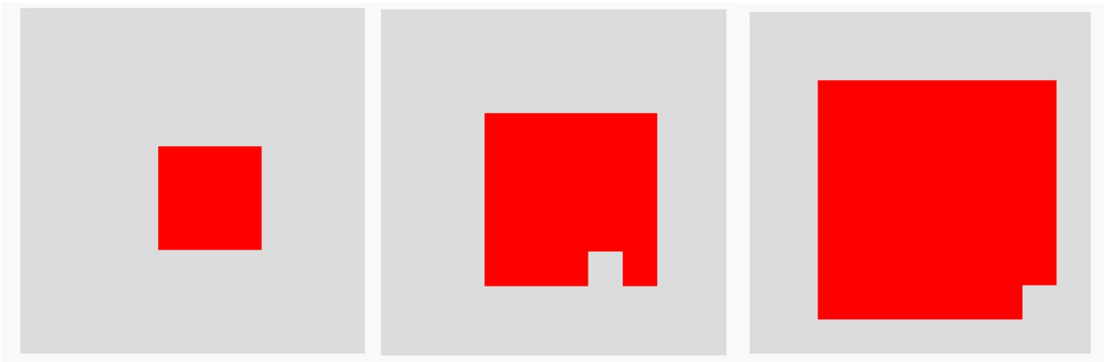
Testing / Experimental Frame

All the experiments were done using a 10x10 grid. The three main parameters to control the testing are Spread probability, forget probability and the neighborhood type. I tested a total of 3 types of spread probability, 0,25, 0.5, and 0.75, I did different neighborhood types of Moore and Von Veumann as well. A couple of tests were also done to ensure that the implementation was correct. First, I did a no spread test where gossip should not propagate beyond the initial source cell. A full spread tests which demonstrate gossip can spread to all reachable cells and become active. A full forgetting test where all cells immediately forget the gossip.

Results

Full Spread Test

Time = 0, 1, 2

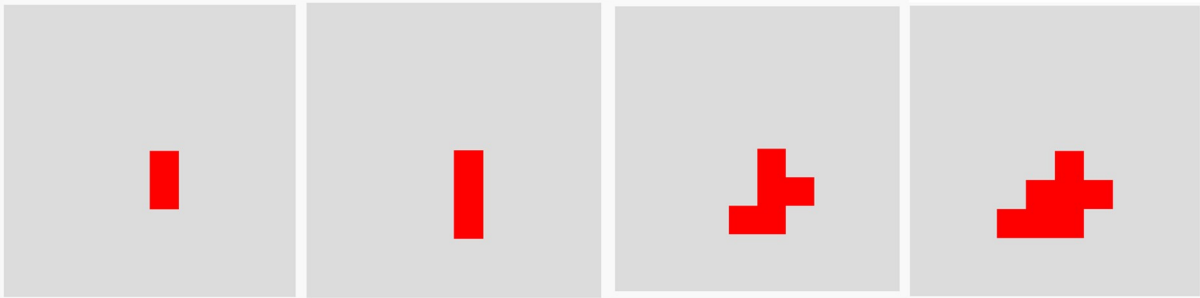


Time = 3, 4

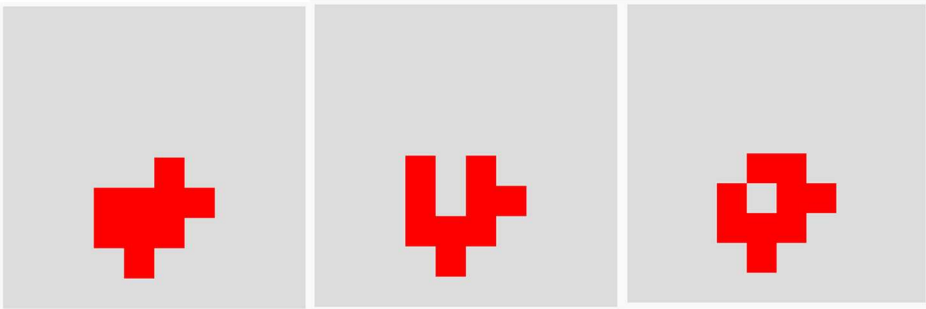
This test simply demonstrates that gossip is able to fully cover the board to ensure the functionality of the model and visualization Json.

Neumann 25% Spread

T = 1, 2, 3, 4



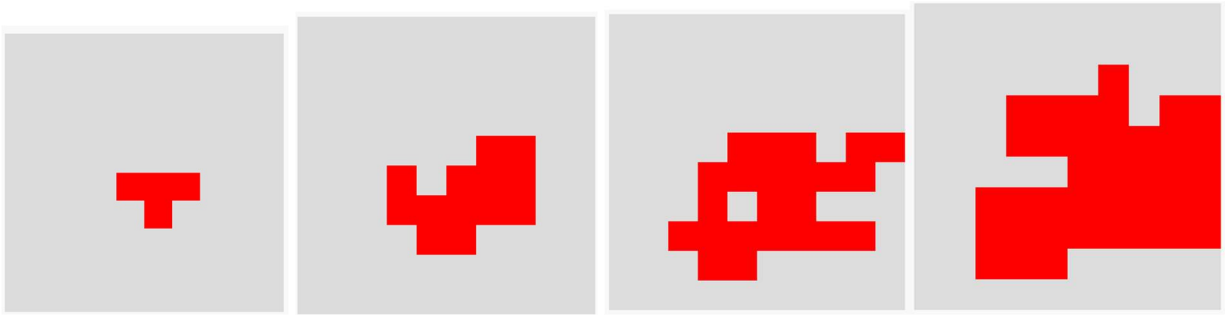
T = 5, 6, 7



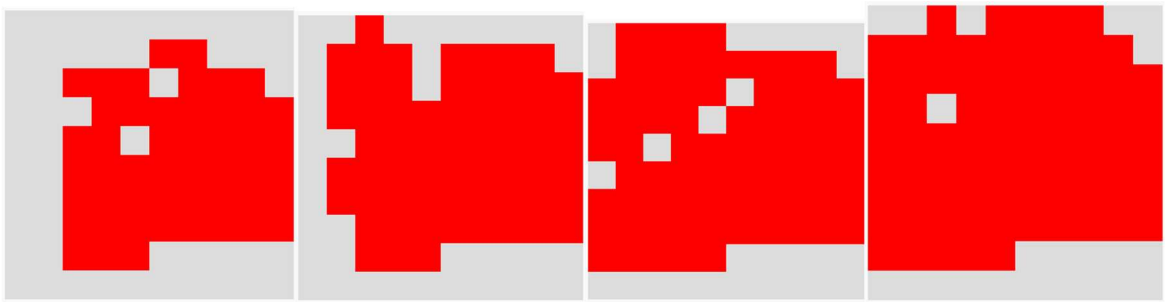
For the Von Neumann with a low spread probability, thew gossip spreads slowly and unevenly. As shown in the images the propagation remains limited to the small region for several time steps. The spread also follows a cross-like pattern which is due to the von Neumann neighborhoods restrictions. The low spread probably results in many failed transmission attempts which is why we see so many gaps.

Neumann 50% Spread

T = 0, 2, 4, 6



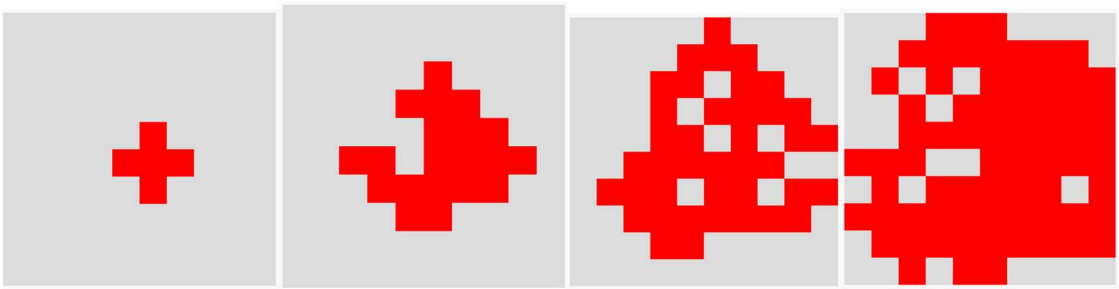
T = 8, 10, 12, 14



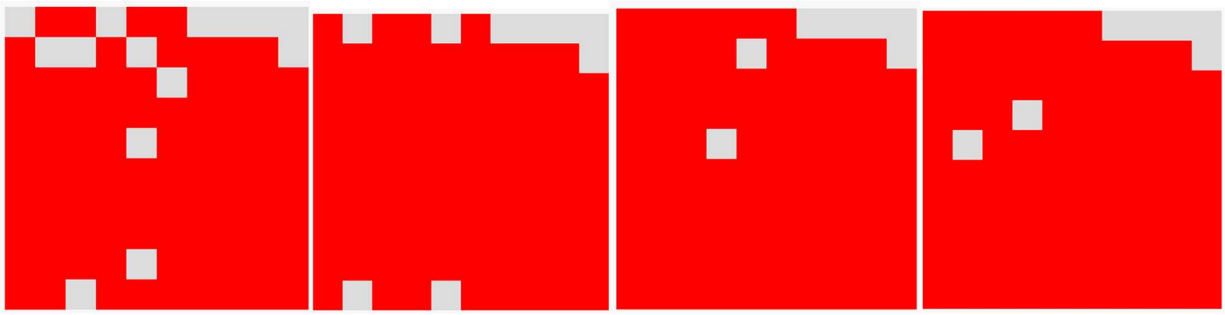
When the spread probability is increased to 50%, gossip propagates more rapidly compared to the 25% case. The images show that the region of active cells grows more quickly and covers a larger portion of the grid over time. Although the spread is faster, the von Neumann still produces the direction patterns and irregular shape for most of the sequence. By the later time stamps most of the grid becomes active indicating that it is enough for the widespread propagation over the whole grid of cells.

Neumann 75% Spread

T = 0, 2, 4, 6



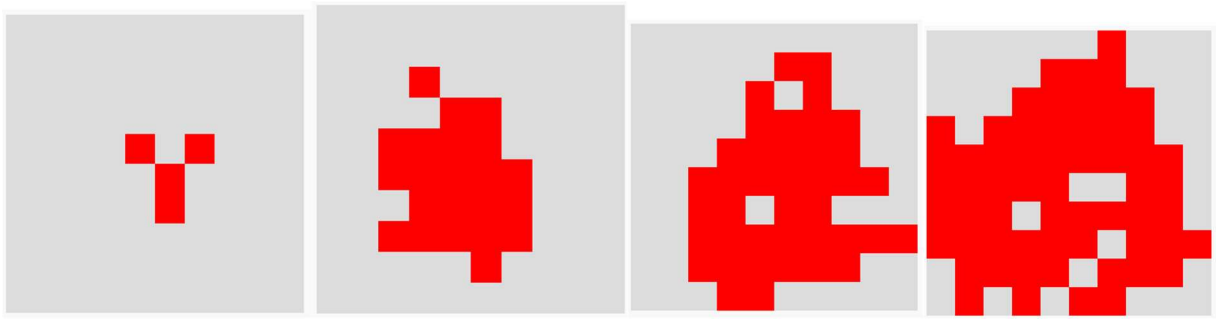
T = 8, 10, 12, 14



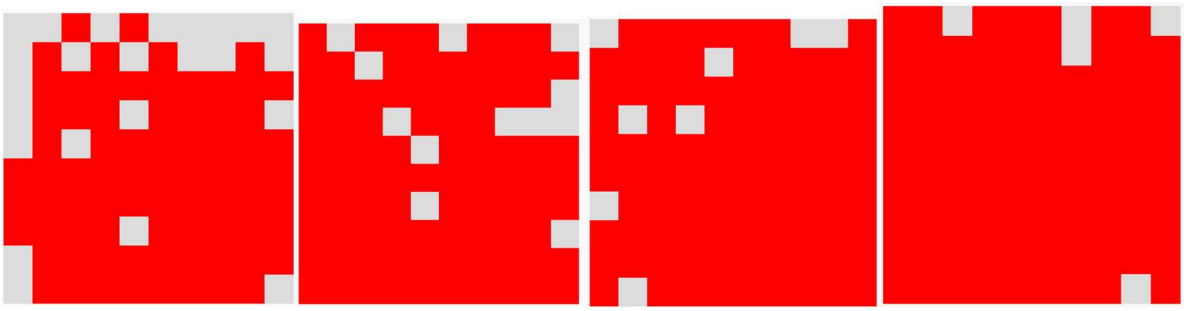
With a high spread probability of 75%, the gossip spreads rapidly across the grid. The images show a much faster expansion compared to the lower probability, with large regions becoming active within only a few time steps. The propagation does still reflect Neumann neighborhoods as seen in the early structure, but by the later time that becomes mostly irrelevant.

Moore 25

T = 0, 2, 4, 6



T = 8, 10, 12, 14

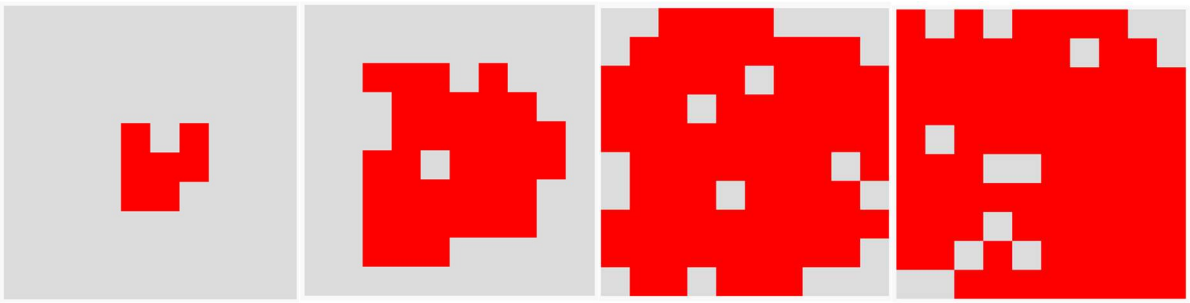


When using the Moore neighborhood with a low spread probability ($p_s = 0.25$), the gossip spreads in all directions due to the increased number of neighbors. Compared to the von Neumann case, the spread appears more circular and less directional.

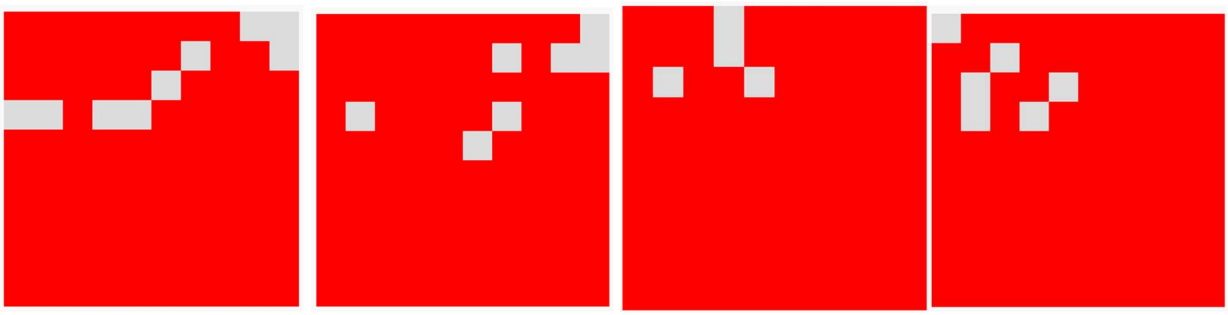
However, the low spread probability still limits propagation, resulting in slow growth and noticeable gaps in the active region. The increased connectivity allows slightly better coverage than the von Neumann case, but the spread remains fragmented

Moore 50

T = 0, 2, 4 ,6



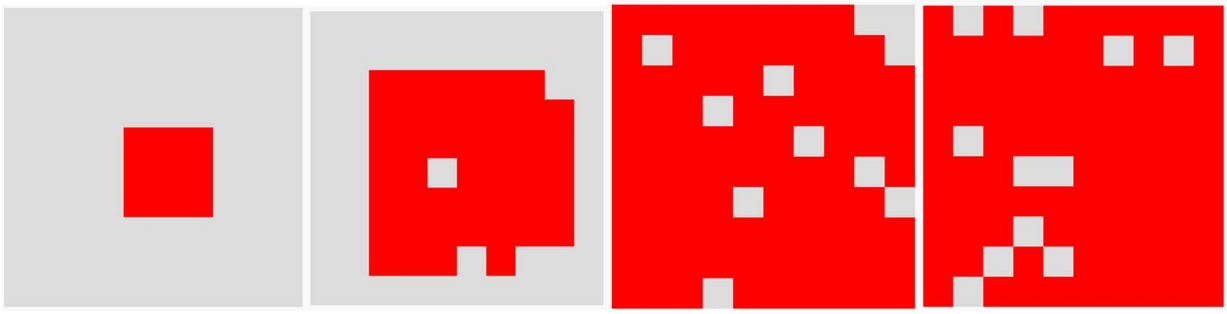
T = 8, 10, 12, 14



At a moderate spread propagation of 50%, the Moore neighborhood produces faster and more uniform propagation compared to the von Neumann case. The images demonstrate how quickly the Moore pattern expansion is. The increased number of neighbors allows gossip to spread more efficiently, filling in the gaps that were present in the lower probability.

Moore 75

T = 0, 2, 4, 6



At a high spread probability of 75%, the Moore neighborhood spreads the gossip rapidly across the grid. Most of the gossip is spread across the grid within a couple time steps. The spread is highly uniform as almost all the neighbors obtain gossip every time step. This shows that the Moore and high spread probability are the most efficient way to distribute gossip.

Overall Observations

Across all experiments, the two key factors are:

- Spread probability: Increasing P leads to a faster and more complete propagation
- Neighborhood structure: Moore neighborhoods result in faster and more uniform spread compared to von Neumann neighborhoods

Additionally, randomness introduces variability in the spread patterns, particularly at lower probabilities, where fragmentation and incomplete propagation are much more common.